Practical 1: Project Initiation, Requirement Analysis, Sitemap, Wireframe, and GitHub Setup

Problem Definition: Initiate the semester-long StudentHub portal by identifying the problemscope , user roles, key modules, navigation flow, and minimum 10 pages.Create a sitemap, low-fidelity wireframe, project folder structure,README file, and GitHub repository.

Objectives

- Develop a centralized portal for students and faculty.
- Manage student academic information.
- Publish announcements and updates.
- Upload assignments and study materials.
- Maintain attendance records.
- Improve student-faculty communication.
- Provide secure role-based login access.

User Roles:

i)Admin

Responsibilities

- Manage students and faculty.
- Manage announcements, assignments, materials, and attendance.
- Control overall portal activities.
- Teacher
- Upload assignments and study materials.
- Post announcements.
- Mark attendance.
- View student information.
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ii) Teacher

Responsibilities

- Login and view profile.
- Read announcements.
- View attendance and assignments.
- Download study materials.
- Functional Requirements

iii) Student

Responsibilities

- Students use the portal to access academic resources and information.
- Responsibilities:
- Register and login securely.
- View personal profile and dashboard.
- Read announcements and updates.
- Check attendance records.
- View assignments.
- Download study materials and lecture notes.

Functional Requirements

User Authentication

- Student registration.
- Secure login and logout.
- Role-based access for Admin, Faculty, and Students.

Student Module

- View dashboard and profile.
- Access announcements.
- View attendance details.
- Download study materials.
- View assigned tasks and assignments.

Faculty Module

- Upload assignments and study resources.
- Post announcements.
- Manage attendance.
- View student details.

Admin Module

- Manage users (students and faculty).
- Manage announcements, assignments, and study materials.
- Manage attendance records.
- Control complete portal functionality.

Non-Functional Requirements

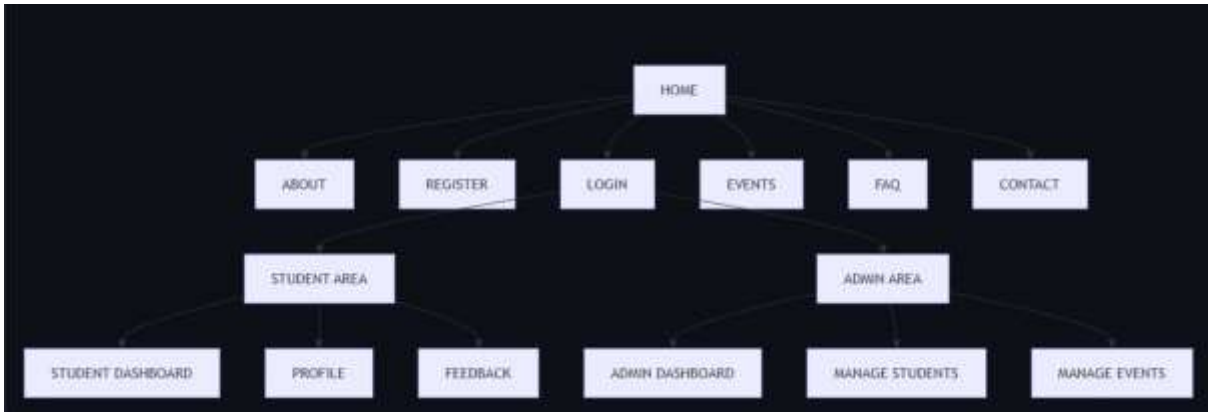
- User-Friendly: Simple interface with easy navigation.
- Security: Secure authentication and protected user data.
- Performance: Fast loading and efficient response time.
- Responsive Design: Accessible on desktop, tablet, and mobile devices.

- Reliability: Accurate storage and management of academic records.
- Maintainability: Easy to update and manage the system.

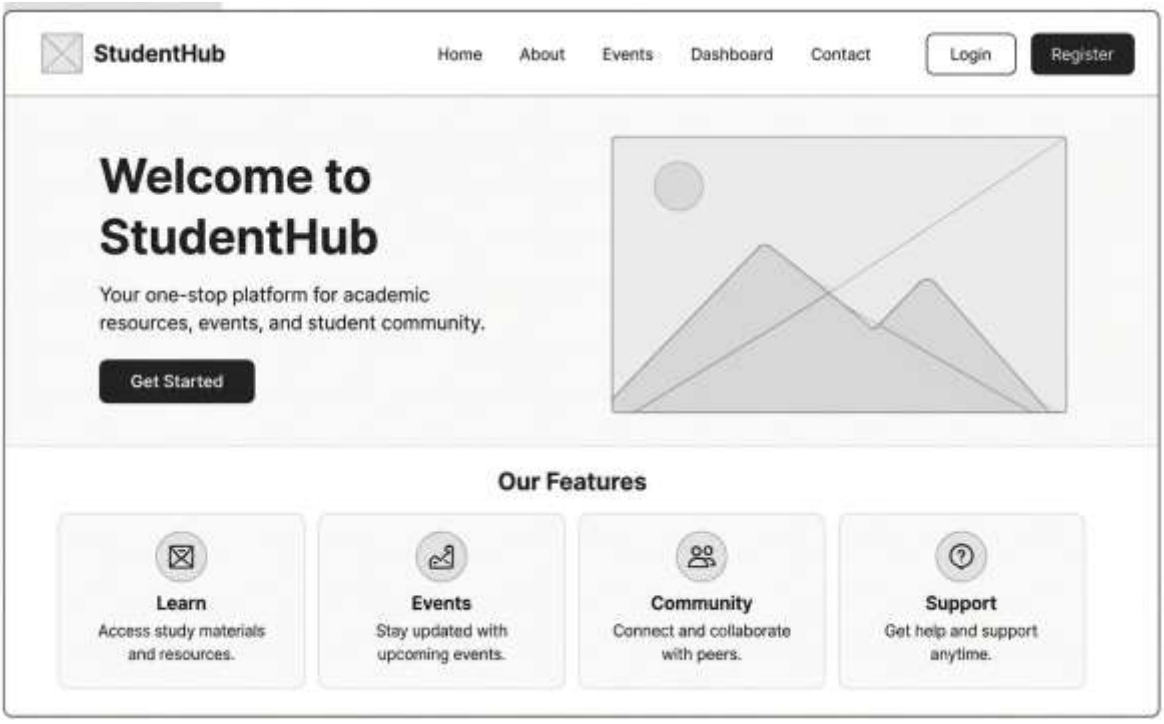
Navigation Flow:



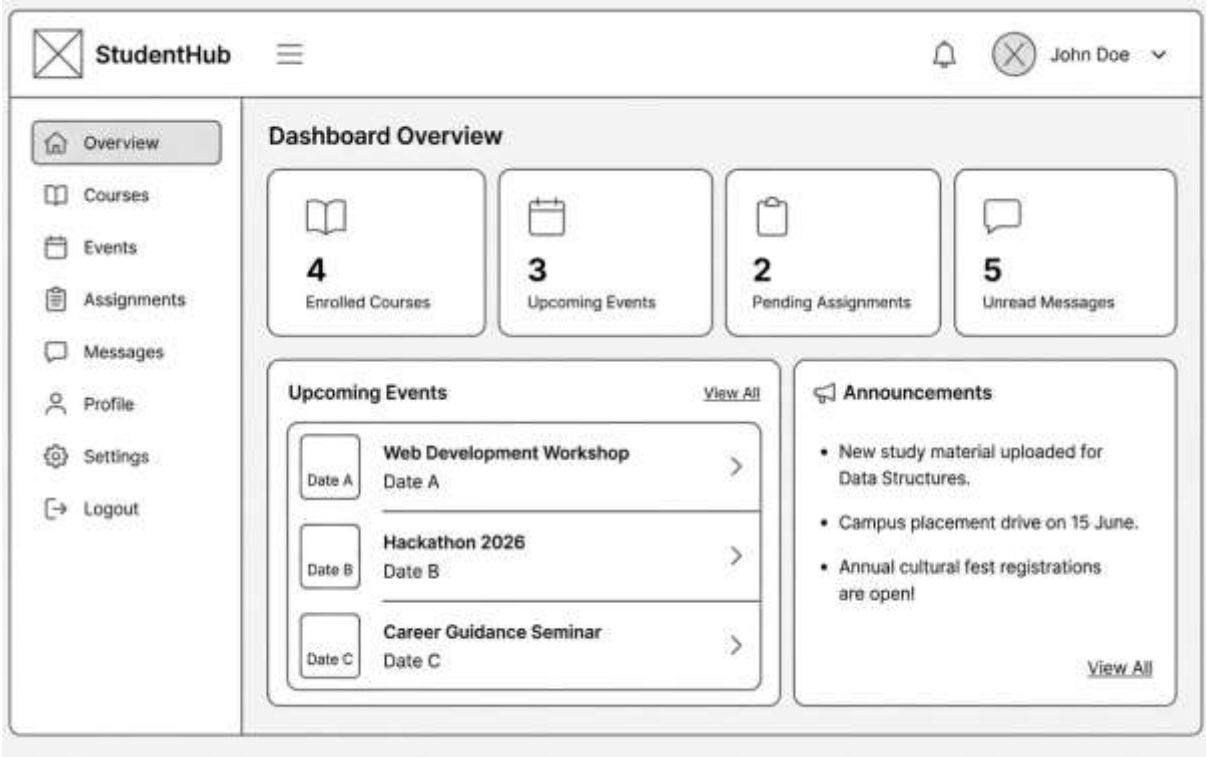
Sitemap:



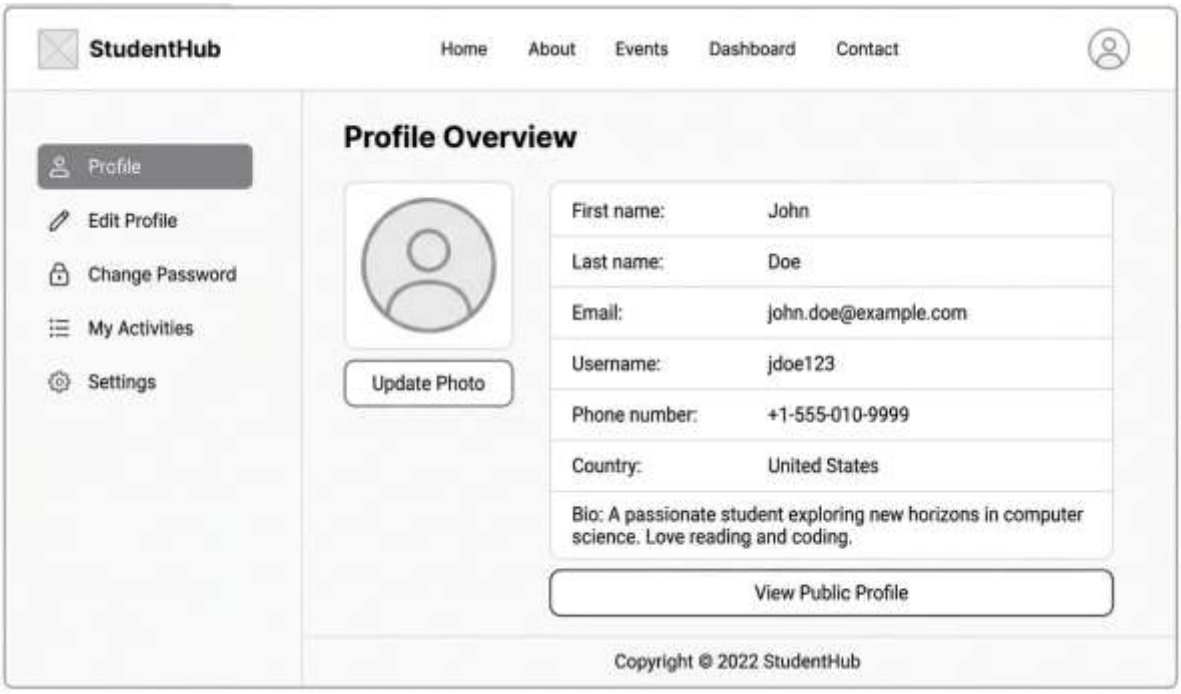
Homepage-wireframe:



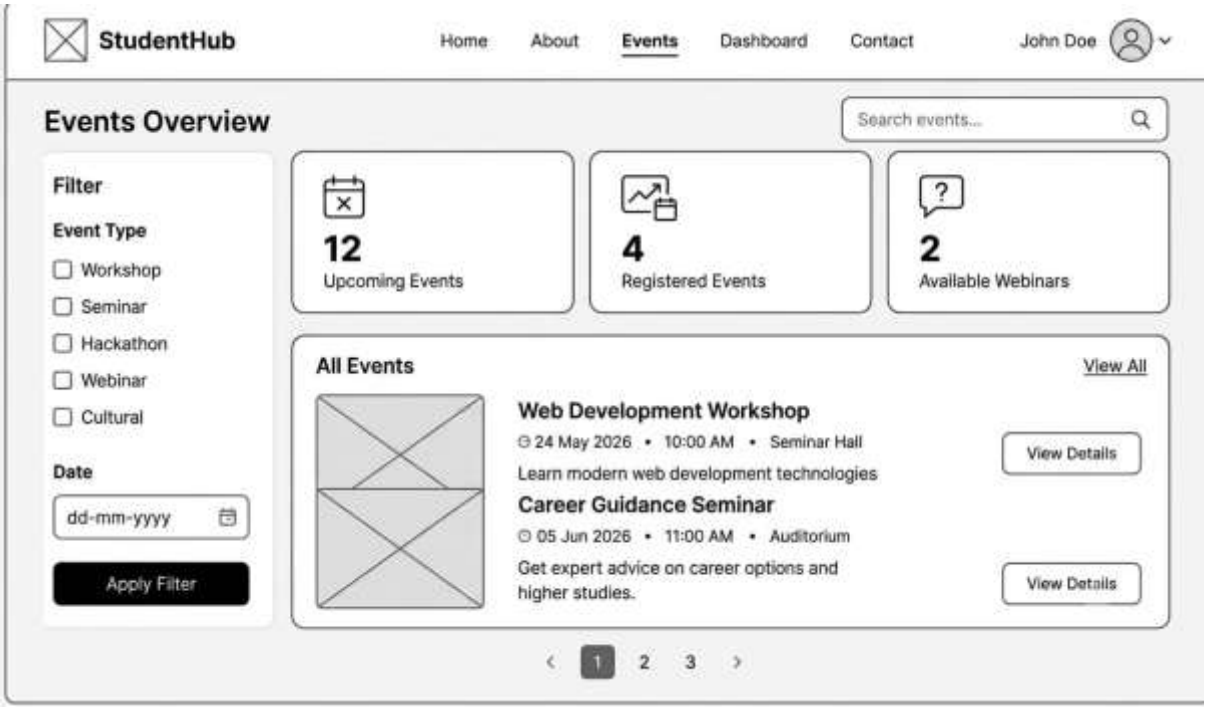
Student Dashbaord-wireframe:



Profile Wireframe:



Event Wireframe:



- **Question-Answer.**
  1. What is a URL and its parts?

     A URL (Uniform Resource Locator) is the web address of a webpage.

     Parts: Protocol (https://), Domain (example.com), Path (/about), Query (?id=1), Fragment (#section).
  2. How does an HTML file process in a web browser?

     The browser reads the HTML file, builds the page structure (DOM), loads CSS and JavaScript, and displays the webpage.
  3. How is page navigation managed among HTML pages?

     Page navigation is managed using HTML links (<a> tags) that connect one HTML page to another.
  4. How are GitHub commits maintained after each practical?

     Create a separate commit after completing each practical with a clear commit message describing the changes.

- **GitHub Repository:**

Link: https://github.com/d26it118-png/Student-Hub.git

Conclusion / Learning Outcome:

The project was successfully initiated by identifying the system requirements, analyzing user needs, preparing a well-structured sitemap, and designing wireframes for the main web pages. A GitHub repository was also created to manage the project efficiently through version control. These initial planning and design activities provide a clear roadmap for the development process and establish a strong foundation for building an organized, user-friendly, and scalable Student Hub website.